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PROJECT TITLE:

ARCHITECTING AND BUILDING LLM-POWERED GENERATIVE AI APPLICATIONS

COURSE TITLE:

CSI3006-SOFT COMPUTING TECHNOLOGIES

TEAM-07

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ABSTRACT:

The huge number of publications in the few years has made it really hard to do a good job of looking at the literature for many subjects. This is because it takes a lot of time to find and look at all the articles for a study. Using keywords to search is not very helpful either. This is because it does not tell the user what the people doing the study were trying to find out. It also does not say what methods they used to do their research and what areas they were looking at. The literature, for disciplines is just too much to handle. Doing an analysis is hard when you have to look at so many publications. The people doing the study used methods and looked at different areas but you cannot tell that from just using keywords. The goal of this proposed project is to develop a tool that will enable scholars and researchers to conduct a thorough literature evaluation through the use of semantic embeddings, which provides a more accurate means of finding and understanding articles that relate to their research area.

This thing uses ideas in big language models and Natural Language Processing methods to give smart analysis of research papers that makes sense. It collects research materials. Gets them ready by uploading documents before pulling out the text. The text it collects is broken down into parts so it can handle a lot of documents without losing the meaning of what the research papers are saying. Research papers are looked at carefully by this method to understand what they are, about. Each section of text is then represented in an electronic format as semantic embedding, where the meaning is inferred rather than by the use of keyword matching.

Using a method called semantic similarity search, we create an embedded query from our user's details and find it in a document embedding database. We then send any associated research data back to our large language model to generate related, coherent responses and provide context through research discoveries, comparative studies, etc., while minimizing the chances of producing false results.

INTRODUCTION:

Finding, reviewing, and summarizing academic literature has altered significantly as a result of the rise of digital research archives. Researchers employ digital technology to assess the literature, identify research gaps, and perform comparative analyses as a result of the increasing number of publications published in journals, conferences, and preprint platforms. Accurate, dependable, and scalable automated systems for literature analysis are desperately needed as the body of scholarly literature and multidisciplinary research expands and changes. Given their impact on scholarly output and research direction, it is imperative that the automated search recommendations are reliable, factual, and contextually relevant.

Recent advances in artificial intelligence, especially with regard to massive language models, present a chance to construct interactive systems that could function as research assistants, gathering data from publications and responding to inquiries about scholarly topics. In contrast to other formal academic resources, probabilistic generation is used in less restricted models of big language systems, which can result in inaccurate summary statements, inaccurate information, and cyclical references.

[1]By using research documents, Retrieval-Augmented Generation (RAG) approaches have been effective in reducing hallucinations and improving the factual accuracy of LLM-generated responses.

[2] Several research studies have been carried out on the application in an effort to expand on the significant efforts to improve the degree of relevance and personalization.

[3] Other research assess the reliability of theoretical systems based on LLM and RAG using each of the post-generation evaluation methodologies. In contrast to the architectural strategies utilized in response production control, the techniques focus on evaluation.

[4] According to the research on knowledge-intensive document analysis, RAG-based systems are more accurate and faithful to the source than LLMs; nonetheless, retrieval is not seen to be enough on its own to prevent unsupported output.

[5] Controlled generation techniques are essential in both scientific and decision support systems since their goal is to produce condition response generation while ensuring that there is enough data recovered that meet this requirement and to have internal mechanisms available to allow for dependable output.

Based on this information, when writing a literature review it is required that the review contain controlled generation techniques such as (i) how much RAG has been able to add to grounding but has not done so at a level significant to support over 50% of conditioned responses; and (ii) LLMs that operate alone (stand alone) tend to be unreliable for providing determinations regarding research papers. In response to these findings, I have completed the RPA assistant (a paper analysis and literature review assistant that uses semantic embedding) in accordance with the above stated requirements

MOTIVATION:

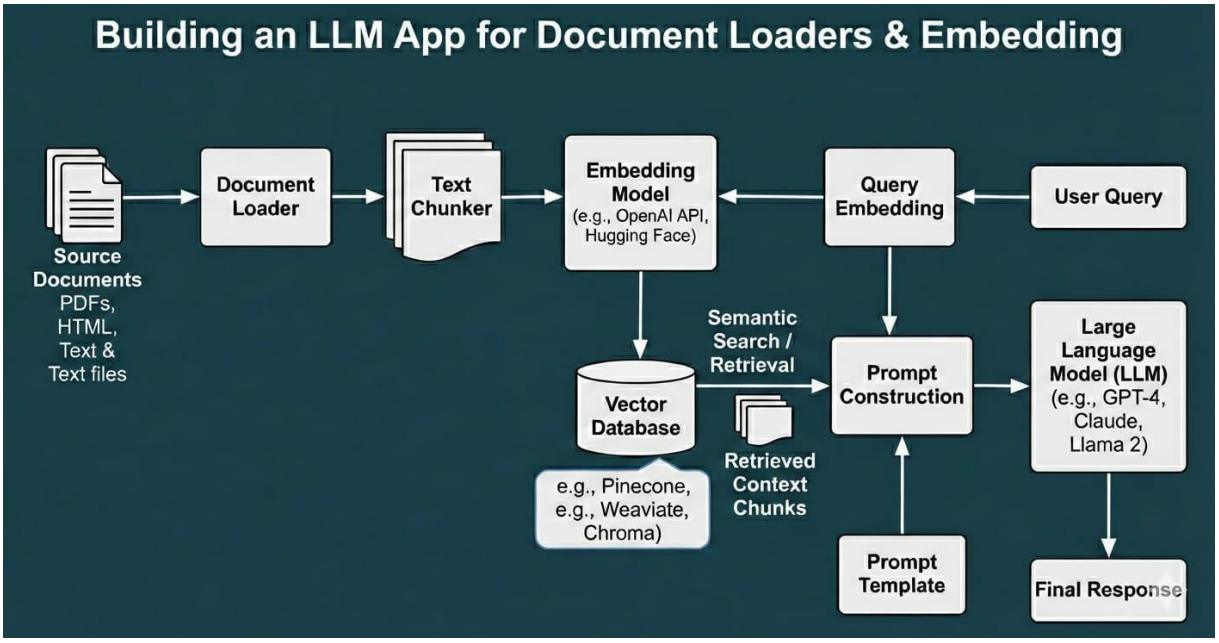
A literature review usually takes a time. This is because many scientific articles are being published every year. When you do research it can be really hard to find the literature in journals that publish a lot of things. The problem is that computers are not good at understanding what research studies really mean. This is because the meaning of research is very complicated and you cannot figure it out by looking at the words that are used in the study. A literature review is still a job and it is hard to find the right literature. Computers are just not able to understand the meaning of research studies, like a literature review. However, by using semantic embeddings, researchers can compare scientific studies using the meaning of the studies instead of comparing them based only on the specific words used to create the study. LLM technology allows researchers to summarize, compare, and evaluate the quality of research articles, which will help them be more creative and efficient with their time.

CHALLENGES:

The size of papers is really big and they are getting more complicated. This means that scientific papers are too long for Large Language Models. To make embeddings for a lot of articles you need a lot of computer power and storage space. It is very hard to find the parts of the paper and if the paper is not broken up correctly you might lose the context that is necessary for understanding the scientific papers. Scientific papers have a lot of information. This information is critical, for understanding the scientific papers. As a result, if there is missing/contextual information an LLM could produce a gibberish/no reasonable summary of that document, thus rendering it useless for use within educational settings.

PAPERS	PROS	CONS
[1] An LLM-Based System for Accessible and Personalized Scientific Communication (Rehman et al., 2025	Accessibility Enhancement, Content Personalization, Audience-Aware Scientific Communication.	Small Sample Size, Limited Experimental Validation, Reduced Generalizability
[2] LLM-Enhanced Bilingual Conversational Agent for EFL Children (Xiao et al., 2025)	Pedagogical Effectiveness, Bilingual Interaction, Child-Centered Language Learning	Age-Restricted Design, Limited Learner Scalability
[3] LLMs and Stack Overflow Discussions (Da Silva et al., 2025)	Empirical Evaluation, Developer Community Insights, LLM Impact Analysis	Platform Dependency, Stack Overflow-Only Scope
[4] Large Language Models Integration in Smart Grids (Madani et al., 2025)	Cross-Domain Innovation, Smart Grid Integration, Intelligent Energy Management	Cybersecurity Vulnerabilities, System Security Risks
[5] Joint Bi-LSTM-GNN for Source Code Retrieval (Bibi et al., 2024)	Hybrid Deep Learning Architecture, Sequential and Structural Feature Learning, Improved Code Retrieval	High Computational Complexity, Increased Training Time
[6] LangurTrace: Forensic Analysis of Local LLM Applications (Jeong et al., 2025)	Digital Forensic Support, Local LLM Traceability, Security Investigation Tool	Tool Dependency, Limited Model Compatibility
[7] LLM-Based Digital Assistant in Assembly Manufacturing (Colabianchi et al., 2024)	Real-World Industrial Deployment, Manufacturing Process Support, Practical LLM Application	Domain Knowledge Dependency, Knowledge Base Quality Sensitivity
[8] Federated and Edge Learning for LLMs (Piccialli et al., 2025)	Privacy-Preserving Learning, Federated Training Framework, Edge-Based Computation	System Coordination Complexity, Resource-Constrained Edge Nodes
[9] LLM-Based Virtual Patient Chatbot (Laverde et al., 2025)	Medical Education Support, Virtual Patient Simulation, Clinical Skill Training	Risk of Inaccurate Responses, Clinical Reliability Concerns
[10] Construction Law Knowledge Repository for LLMs (Zhou et al., 2025)	Domain-Specific Knowledge Integration, Legal Information Structuring, Enhanced LLM Performance	Narrow Domain Applicability, Jurisdictional Limitations

ARCHITECTURE DIAGRAM:



LITERATURE SURVEY:

[1] Rehman, M., Petrillo, A., Awasare, K., "*An LLM-Based System for Accessible and Personalized Scientific Communication,*" Procedia Computer Science, vol. 274, pp. 939–952 (or similar), Elsevier, 2025

Reference Link: https://doi.org/10.1016/j.procs.2025.12.092

[2] Xiao, F., Li, Z., Lin, J., Zou, X., Yang, D., Zou, W., & Xiong, J., "*Leveraging an LLM-enhanced bilingual conversational agent for EFL children's dialogic reading: Insights from children, parents, and educators*," Computers and Education: Artificial Intelligence, vol. 9, p. 100484, Elsevier, 2025.

Reference Link: https://doi.org/10.1016/j.caeai.2025.100484

[3] Da Silva, L. M. P., Samhi, J., & Khomh, F., "*LLMs and Stack Overflow discussions: Reliability, impact, and challenges*," Journal of Systems and Software, vol. 230, p. 112541, Elsevier, 2025.

Reference Link: https://doi.org/10.1016/j.jss.2025.112541

[4] Madani, S., Tavasoli, A., Khoshtarash Astaneh, Z., & Pineau, P.-O., "*Large Language Models Integration in Smart Grids*," Energy Reports, vol. 14, pp. 1562–1577, Elsevier, 2025.

Reference Link: https://doi.org/10.1016/j.egyr.2025.06.051

[5] Bibi, N., Maqbool, A., & Rana, T., "*Enhancing source code retrieval with joint Bi-LSTM-GNN architecture: A comparative study with ChatGPT-LLM*," Journal of King Saud University - Computer and Information Sciences, vol. 36, p. 101865, Elsevier/King Saud University, 2024.

Reference Link: https://doi.org/10.1016/j.jksuci.2023.101865

[6] Jeong, S., Lee, S., & Park, J., "*LangurTrace: Forensic analysis of local LLM applications,*" Forensic Science International: Digital Investigation, vol. 54, Article 301987, Elsevier, 2025.

Reference Link: https://doi.org/10.1016/j.fsidi.2025.301987

[7] Colabianchi, S., Costantino, F., & Sabetta, N., "*Assessment of a large language model based digital intelligent assistant in assembly manufacturing,*" Computers in Industry, vol. 162, Article 104129, Elsevier, 2024.

Reference Link: https://doi.org/10.1016/j.compind.2024.104129

[8] Piccialli, F., Chiaro, D., Qi, P., Bellandi, V., & Damiani, E., "*Federated and edge learning for large language models*," Information Fusion, vol. 117, Article 102840, Elsevier, 2025.

Reference Link: https://doi.org/10.1016/j.inffus.2024.102840

[9] Laverde, N., Grévisse, C., Jaramillo, S., & Manrique, R., "*Integrating large language model-based agents into a virtual patient chatbot for clinical anamnesis training*," Computational and Structural Biotechnology Journal, vol. 27, pp. 2481–2491, Elsevier, 2025.

Reference Link: https://doi.org/10.1016/j.csbj.2025.05.025

[10] Zhou, S., Wang, H., Ng, S. T., Li, D., Xie, S., Chen, K., & Wang, W., "*Building a construction law knowledge repository to enhance general-purpose large language model performance on domain question-answering: A Chinese case study*," Engineering, Construction and Architectural Management, accepted 2025.

Reference Link: https://doi.org/10.1108/ECAM-06-2024-0701